
SUMMARY

Versatile Software Engineer with a strong foundation in AI/ML development, web development, and cybersecurity. Author of peer-reviewed research with Comcast and Amazon, demonstrating innovation in optimising network performance using advanced algorithms and machine learning.

EXPERIENCE

ITMagnet

Apr 2025 – Present

Software Developer

Sydney, AU

- Shipped features across the **Onsite ERP** suite, supporting internal operational workflows and client-facing business platforms.
- Built an internal **Purchase Requisition** system along with a team that streamlined approval workflows and improved procurement tracking and transparency.
- Delivered the **ApplyEasy** integration with OnsiteLive for secure, seamless data exchange between external apps and enterprise systems, in this case, an external form filler application.
- Designed a **Headless CMS** dashboard (Builder.io), enabling flexible content management and faster UI iteration cycles.
- Maintain and operate **Azure** cloud infrastructure powering Onsite Services' production and internal platforms.
- Ran structured evaluation of **IoT tracking devices** (IoTTag, Netstar) for large-scale mining asset tracking — telemetry accuracy, GPS reliability, battery, and API feasibility — to inform client deployment strategy.
- Led integration of **Large Language Models (LLMs)** into production workflows — prompt engineering and AI-driven validation/chatbot systems — and re-architected the AI-powered **Onsite Quote Validation** engine with NLP to improve interpretation accuracy and automated decision support.

Deakin University

Apr 2025 – Oct 2025

Associate Research Fellow

Melbourne, AU

- **SmartSat CRC (Defence Project)** — developed intelligent **dynamic spectrum allocation (DSA)** algorithms using **Deep Q-Networks (DQN)**, **Asynchronous Advantage Actor-Critic (A2C)**, and LLMs to enable opportunistic reuse of underutilised GEO satellite bands by LEO satellite systems.
- Drove interdisciplinary research across satellite communications, AI, and signal processing to enhance spectrum efficiency and satellite network performance.

Deakin University

May 2024 – Apr 2025

Research Assistant

Melbourne, AU

- Leveraged **LLMs** (Llama3 / GPT-2 / T5) to predict congestion events in the FreeBSD network stack, boosting **L4S** performance; built a standalone **AQM-LLM** platform.
- Designed and implemented a modified **L4S Active Queue Management (AQM)** algorithm in the FreeBSD kernel in **C**, and applied **asynchronous gradient descent (A3C)** to refine buffer management — achieving **sub-5ms latency at the 99th percentile**.
- Delivered an industry-funded project for **Comcast**, combining **DDPG** with online convex optimisation to improve fairness and performance in multipath congestion control; built Linux kernel implementation and Bash automation benchmarking **DDPG-MPCC** against state-of-the-art **MPTCP**.
- Led a feasibility study pairing **RF sensing** with **Correlated Knowledge Distillation (CKD)** for real-time human pose monitoring; built teacher/student **ResNet**-based fully convolutional models in **PyTorch** and **OpenCV** (AlphaPose).
- Managed operations at Deakin's IoT & SE lab, mentoring students through final-year research.

Larson & Turbo Technology Services

Oct 2023 – Jan 2024

Cybersecurity Engineer

Pune, IN

- Provided **client-facing security consulting** to **Honeywell** as part of a collaborative team.
- Conducted **risk analysis** and threat identification with **vulnerability scanning** tools, prioritising findings by **CVE score**.
- Hardened Linux-embedded devices by remediating **system vulnerabilities**, updating core packages, and applying patches.

TECHNICAL SKILLS

Languages: C, C#, Python, Bash, Java, JavaScript, TypeScript

AI / ML & Data Science: LLMs, Hugging Face, scikit-learn, TensorFlow, PyTorch, OpenAI Gym, OpenCV, Spark, Hadoop, Statistics, Data Wrangling, Time-Series (ARIMA)

Web & Backend: .NET 10, Node.js (Express), Angular, React, Spring Boot, Maven, REST APIs, Apache, HTML

Cloud & DevOps: Microsoft Azure, AWS (EC2), Git, Docker

Security: Vulnerability Scanning, Risk Assessment & Mitigation, TCP/IP & AQM, Network Traffic Monitoring, Patching, Kernel Programming

PROJECTS

Smart Lighting System [Demo](#)

- Architected a scalable system for user management and device control, deployed on **AWS EC2** with load balancers to sustain high traffic.
- Built a **microservices** architecture (JavaScript, AngularJS, Node.js, MongoDB) letting users register lighting devices and control settings via **MQTT**.

Baby Monitoring System [Code](#)

- Led scrum-based development of an IoT web app that monitors infants and pushes critical alerts via **IFTTT** triggers and emergency email.
- Built with **Express.js** REST APIs, **Passport.js** auth, and AngularJS; tested with Karma, Jasmine, and Protractor (Selenium WebDriver); styled with Bootstrap and Semantic UI.

EDUCATION

Deakin University

*B.E. (Honours), Software Engineering — **First Class Honours***

2019 – 2023

Melbourne, Victoria

PUBLICATIONS

1. **Distilling Large Language Models for Network Active Queue Management.** Pokhrel, S.R.; **Satish, D.**; Kua, J.; Walid, A. *IEEE Transactions on Networking* 2026, 34, 5501–5513. [doi:10.1109/TON.2026.3690076](#)
2. **Active Queue Management in L4S with Asynchronous Advantage Actor-Critic: A FreeBSD Networking Stack Perspective.** **Satish, D.**; Kua, J.; Pokhrel, S.R. *Future Internet* 2024, 16(8), 265. [doi:10.3390/fi16080265](#)
3. **DDPG-MPCC: An Experience Driven Multipath Performance Oriented Congestion Control.** Pokhrel, S.R.; Kua, J.; **Satish, D.**; Ozer, S.; Howe, J.; Walid, A. *Future Internet* 2024, 16(2), 37. [doi:10.3390/fi16020037](#)
4. **Deakin RF-sensing: Experiments on Correlated Knowledge Distillation for Monitoring Human Postures with Radios.** Pokhrel, S.R.; Kua, J.; **Satish, D.**; Williams, P.; Zaslavsky, A.; Loke, S.W.; Choi, J. *IEEE Sensors Journal* 23(22), 28399–28410. [doi:10.1109/jsen.2023.3320131](#)